

Spectroscopy of unbound ^{17}C states

via $^{16}\text{C}(\text{d},\text{p})^{17}\text{C}$ at FRIB (a1975)

Angular distributions · absolute normalization · L assignments

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spectroscopic reference: Movilla & Ayyad (2026-02)

- ^{16}C beam in the AT-TPC, D_2 active target (FRIB a1975).
- 188.33 MeV total = 11.77 MeV/u; $Q_{g.s.} = -1.498$ MeV.
- Transfer populates the bound region + seven unbound resonances.
- Goal: per-state $d\sigma/d\Omega$ on an absolute scale
 - $\rightarrow$ transferred- L from FRESKO ADWA shape comparison.
- Data reduced with ATTPCROOT (InterpSolver); cont. of C16dp_fits.

Unbound resonances fitted

Ex (MeV)	Jpi	Gamma (keV)

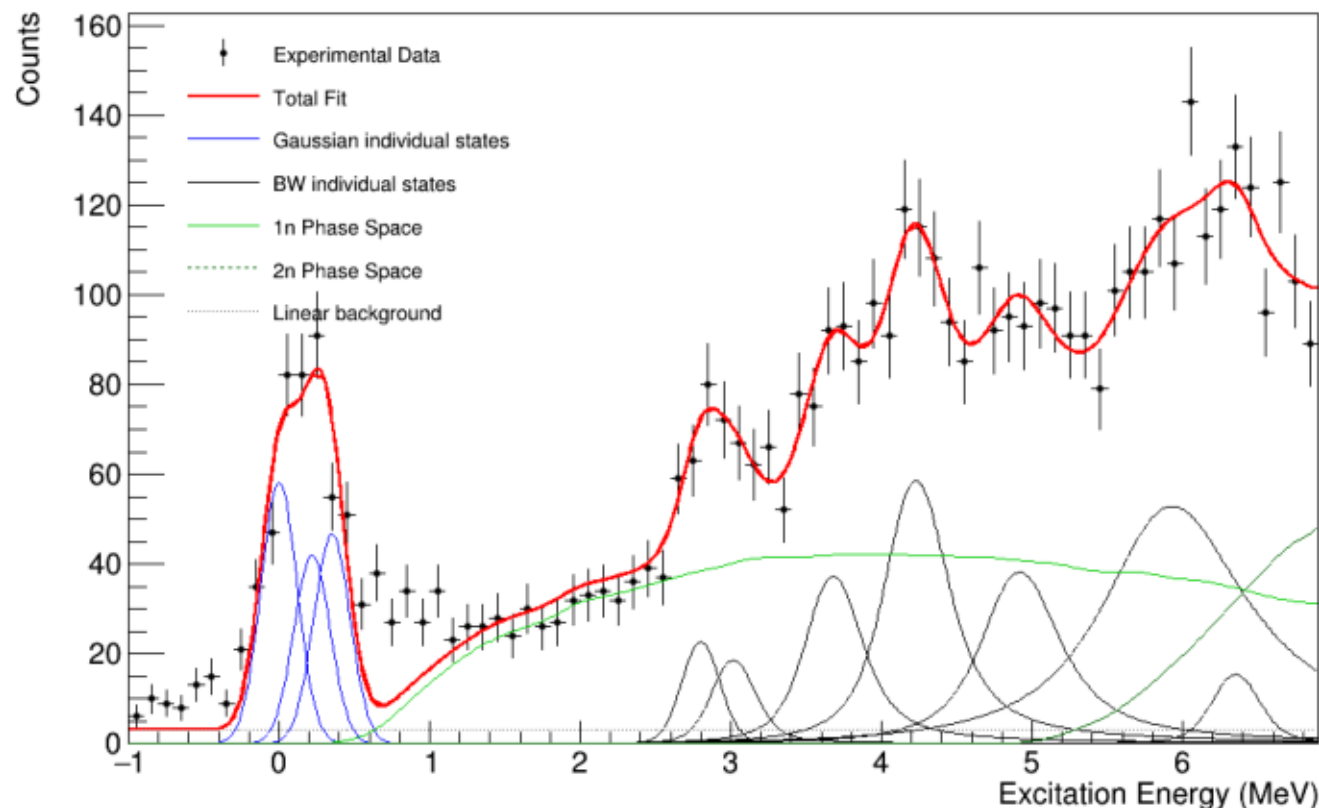
2.763	1/2-	50
2.980	3/2+	200
3.661	new	398
4.231	3/2+	500
4.841	1/2+	300
5.91	3/2+	1500
6.30	5/2+	396

- 1. Spectral fit (C16_pd_AngBins.C): $12 \times 2.5^\circ$ bins, $10\text{--}40^\circ$.
 - 3 Gaussians (bound) + 7 penetrability-modified Breit-Wigners
 - + $1n/2n$ phase space + linear bg; analytic Voigt lineshape.
 - Inclusive fit pins positions/widths; per-bin frees amplitudes.
- 2. Efficiency: 8 Ex-tables, bilinear (E_x, θ); floor+gradient cuts.
- 3. Absolute normalization fixed by the $^{16}\text{C}(d,d)$ elastic channel.
- 4. L scan: each AD vs FRESCO $L=0\text{--}3$ ADWA shapes; best L by χ^2/ν .
- 5. Systematics: 82-variant campaign $\rightarrow$ per-state envelope.

Inclusive E_x spectrum: the global fit

spectral fit

Inclusive 10-40 deg



- A localized $+2\text{--}4\sigma$ excess near 0.4–1.1 MeV is not absorbed by the nominal model.

- Adding a contaminant Gaussian (centroid pegs ~ 0.45 MeV) restores an acceptable fit:

cuts	nominal	+contam

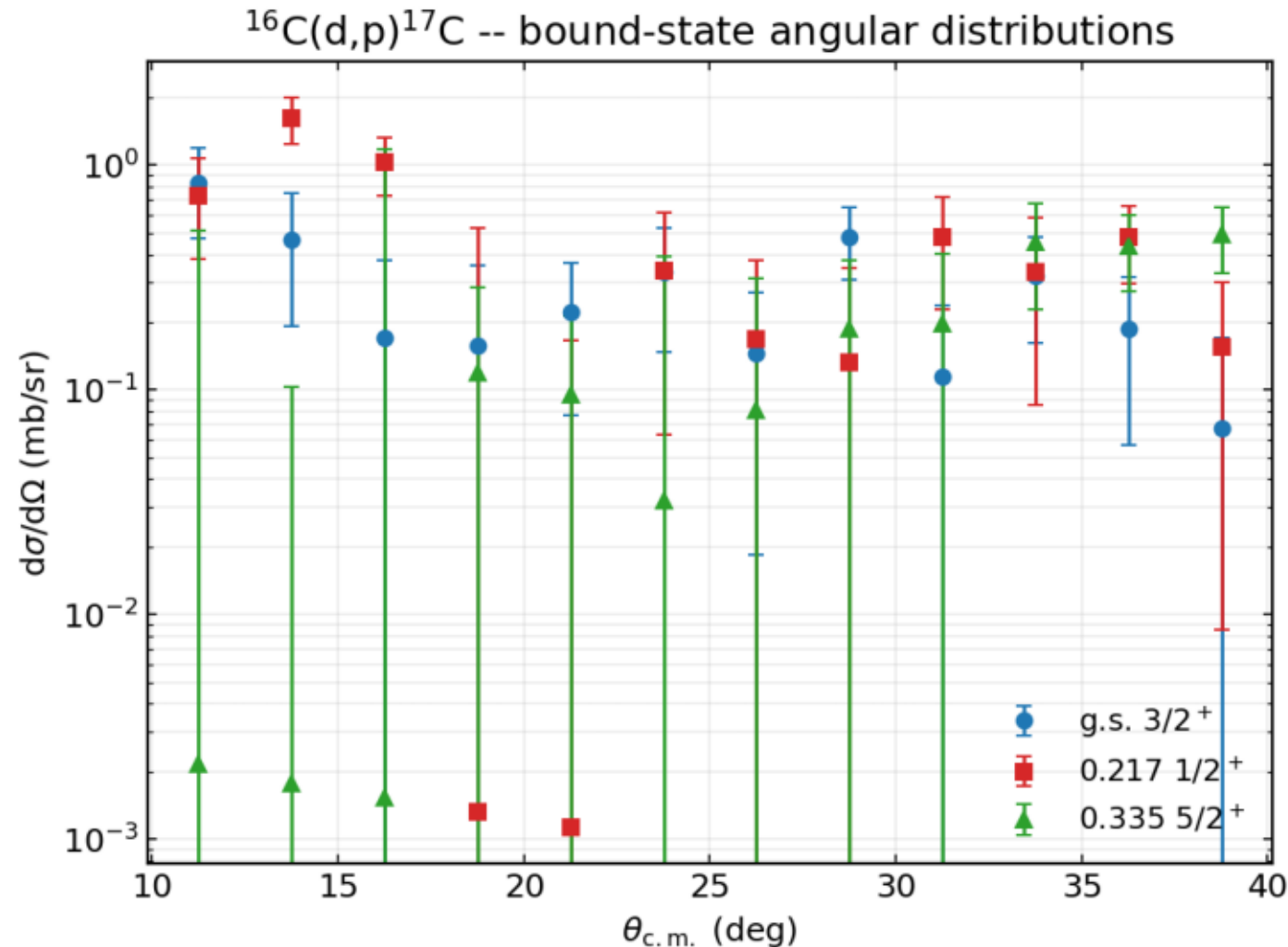
loose	1.78	0.84
strict	1.88	0.97
	(chi2/nu)	

- Survives strict cuts (0.450 $\rightarrow$ 0.451 MeV) $\Rightarrow$ not an AT-TPC edge artifact.

- Carried as a model-dependence systematic.

Bound states: angular distributions

comparison



- Three bound ^{17}C states, as a cross-check of the machinery:

- g.s. $3/2^+$

- 0.217 MeV $1/2^+$

- 0.335 MeV $5/2^+$

- Forward-peaked g.s. and 0.217 shapes consistent with the expected ℓ content.

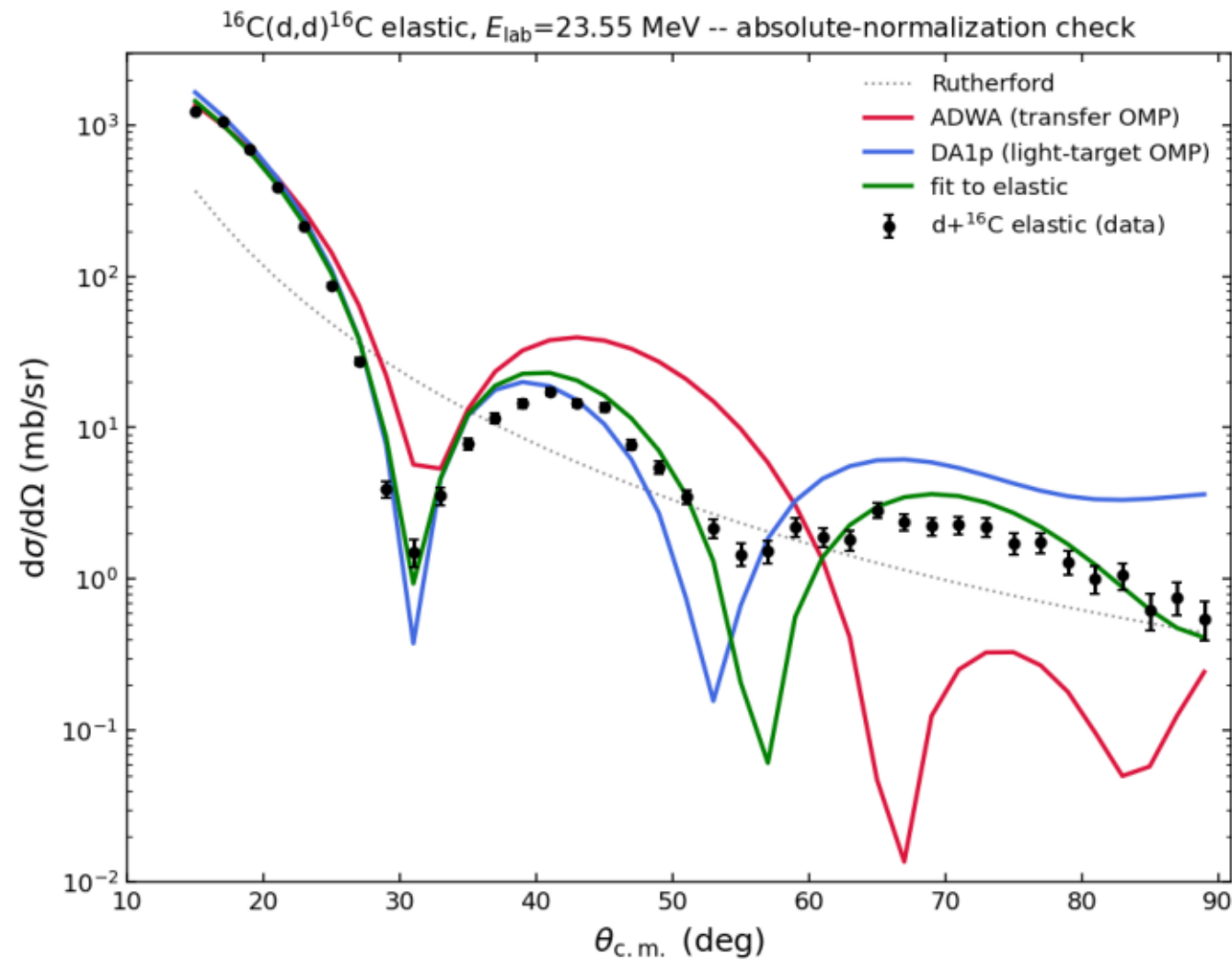
- The $5/2^+$ is only weakly populated — its points sit near the detection floor (large bars), shown for completeness.

- Validate efficiency + absolute scale before the unbound region.

Counts → cross section (fixed luminosity constant):

$$\frac{d\sigma}{d\Omega} = \frac{Y}{N_{beam} N_{tgt} \Delta\Omega} \cdot 10 / \varepsilon$$

- $N_{beam} = 1.6146 \times 10^5$, $N_{tgt} = 0.019632$, $\Delta\Omega = 2\pi(\cos \theta_{lo} - \cos \theta_{hi})$.
- Cross-check: the $^{16}\text{C}(\text{d},\text{d})$ elastic channel shares the entrance partition (same beam, D_2 target, luminosity).
 - An absolute OM elastic cross section that reproduces the data validates the constant — no free normalization.
- Self-contained partial-wave OM solver (om_elastic.py): Numerov + Coulomb matching; vs Rutherford to 10^{-3} .
 - $E_{lab}(\text{d}) = 23.55 \text{ MeV}$, $E_{c.m.} = 20.92 \text{ MeV}$.



Forward-lobe (data/OM):

OMP	chi2/nu	d/OM

ADWA (transfer)	398	0.864
DA1p (global)	105	0.886
free 5-par fit	32	0.978

- Absolute scale consistent with unity — no rescale.
- BUT the transfer ADWA potential itself sits ~14% low.
- Honest figure = cross-potential spread $\approx 0.86\text{--}0.98$ (~12%), a GLOBAL scale uncertainty (not in per-point errors).

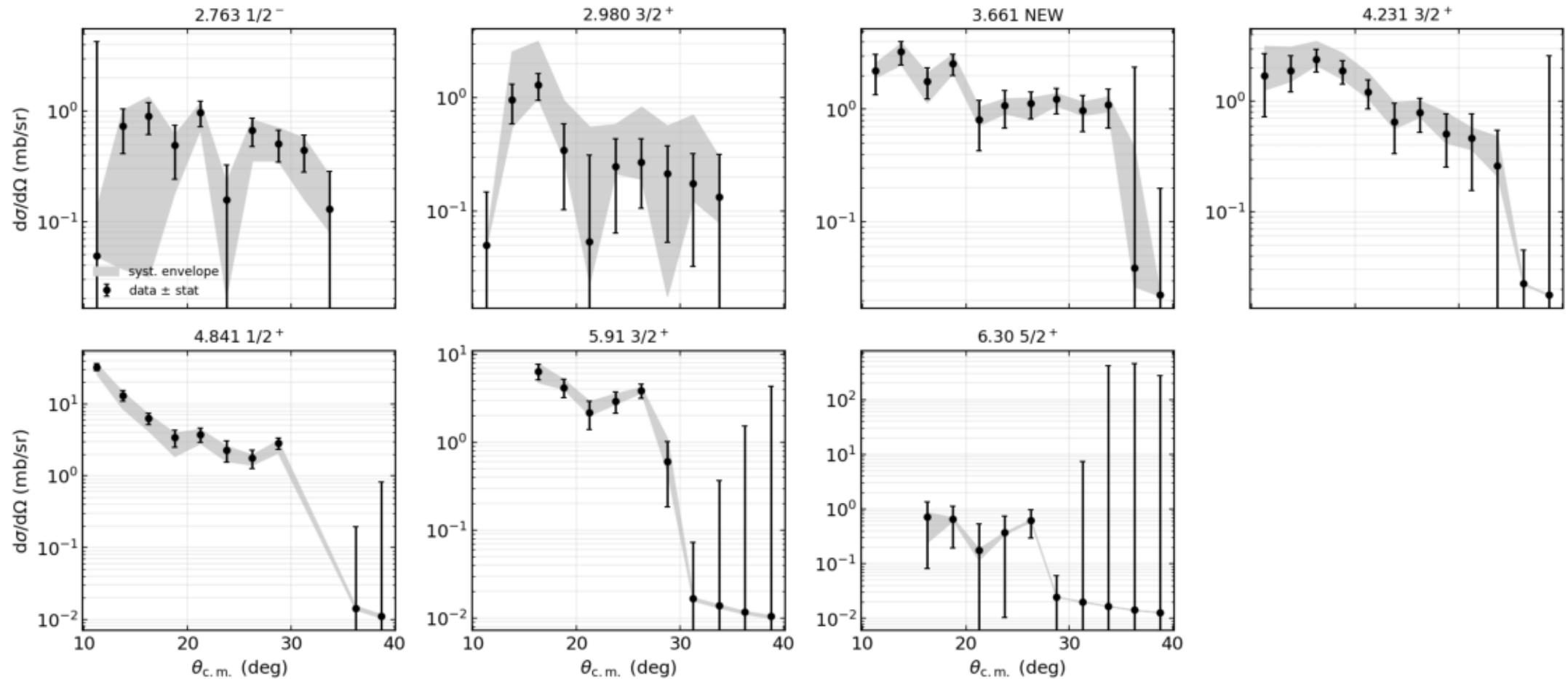
$$\delta\left(\frac{d\sigma}{d\Omega}\right) = \frac{d\sigma}{d\Omega} \sqrt{(\delta Y/Y)^2 + (\delta\epsilon/\epsilon)^2}$$

- δY = Minuit amplitude error (χ^2 fit to $\sqrt{N}$ -weighted histograms) — already carries Poisson stats.
- FIXED (2026-05-30): a previous $\sqrt{1/Y + (\delta Y/Y)^2}$ double-counted Poisson ($\sqrt{2}$ inflation).
 - e.g. g.s. 10-12.5°: 0.433 → 0.356 mb/sr.
- Three uncertainty layers, kept separate:
 - statistical (above) + systematic (82-variant envelope) + global scale (~12% OMP spread).

Unbound resonances: $d\sigma/d\Omega$ + systematic band

angular distributions

$^{16}\text{C}(d,p)^{17}\text{C}$ unbound resonances -- $d\sigma/d\Omega$ with 82-variant systematic band



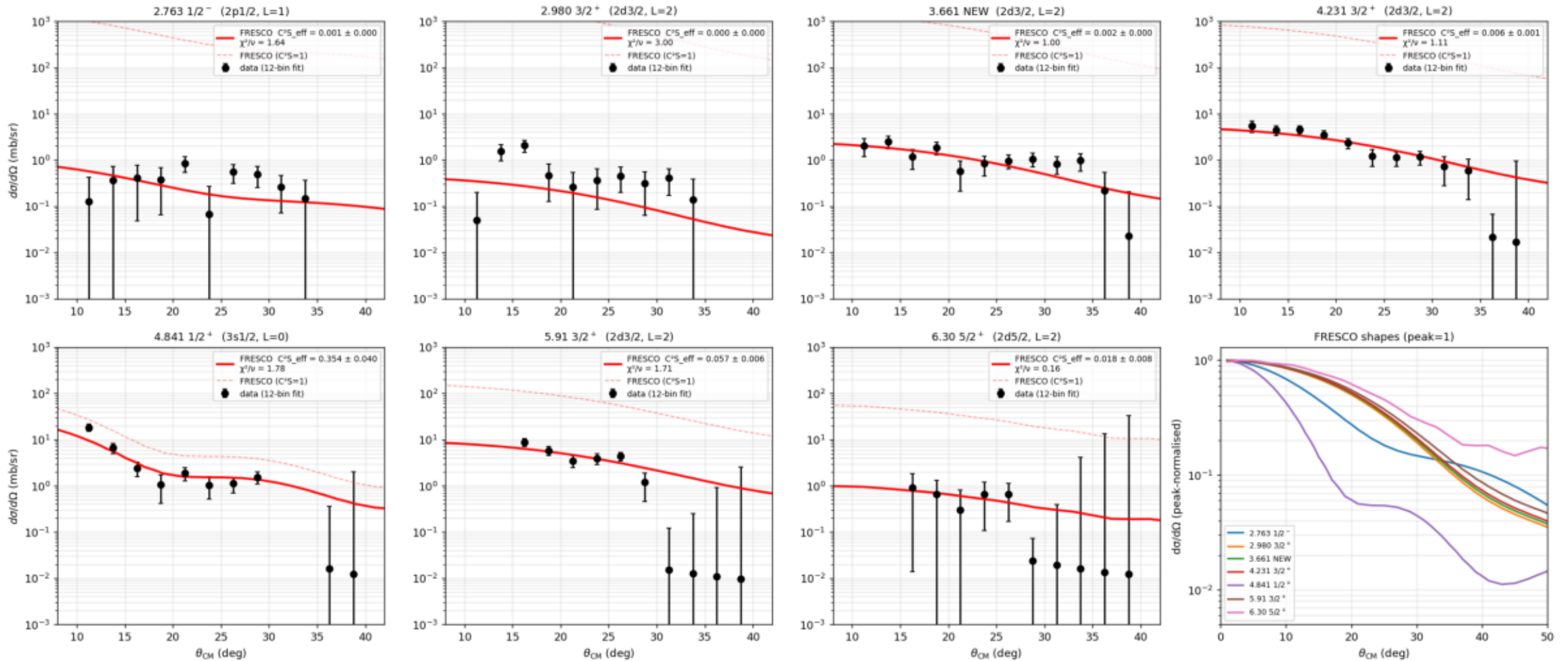
Black: data $\pm$ corrected statistical error. Gray: min-max across the 82 spectral-fit variants. Fits use $\theta_{c.m.} \leq 30^\circ$.

- FRESCO 7-state DWBA with the Johnson–Soper ADWA effective deuteron OMP (sum of Koning–Delaroche $p + {}^{16}\text{C}$ and $n + {}^{16}\text{C}$ at $E_d/2$).
- Four templates $L=0,1,2,3$ (neutron in $3s_{1/2}, 2p_{1/2}, 2d_{3/2}/2d_{5/2}, 1f_{5/2}$); weakly-bound approx $b_e=0.5$ MeV.
- Each AD fit to each L shape (single normalization); best L by χ^2/ν on $\theta_{c.m.} \leq 30^\circ$.
- SHAPE-ONLY: the WBA reproduces the L-dependent shape but not the magnitude $\Rightarrow C^2S$ are effective, not publication SFs
 - (needs CDCC / pole-residue treatment).

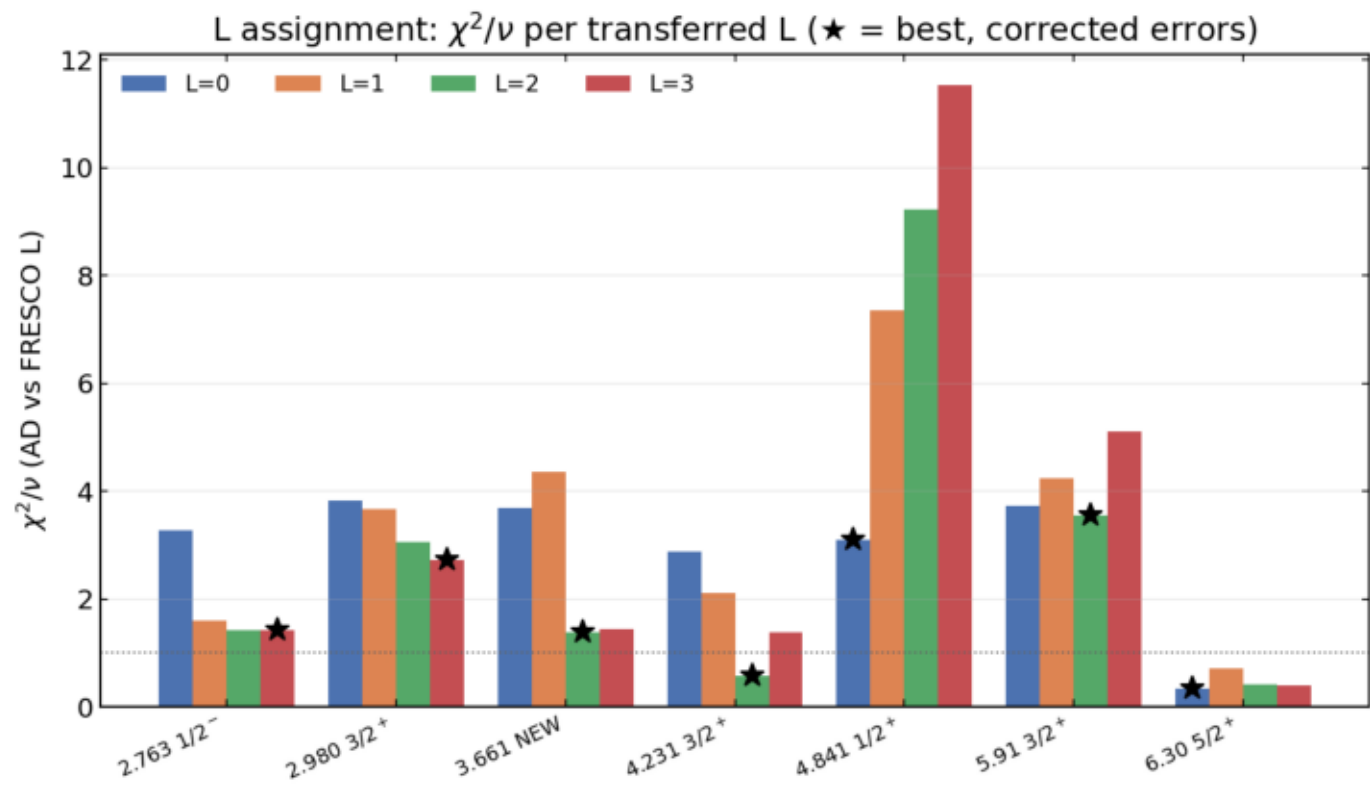
L assignment: FRESCO shape overlays

L values

16C(d,p)17C @ 11.77 MeV/u — FRESCO ADWA (weakly-bound approx) vs extracted AD



Curves = FRESCO ADWA at each state's literature single-particle config (normalization-fit, full 10–40°). The data-driven all- L scan on $\theta_{c.m.} \leq 30^\circ$ is the next slide.



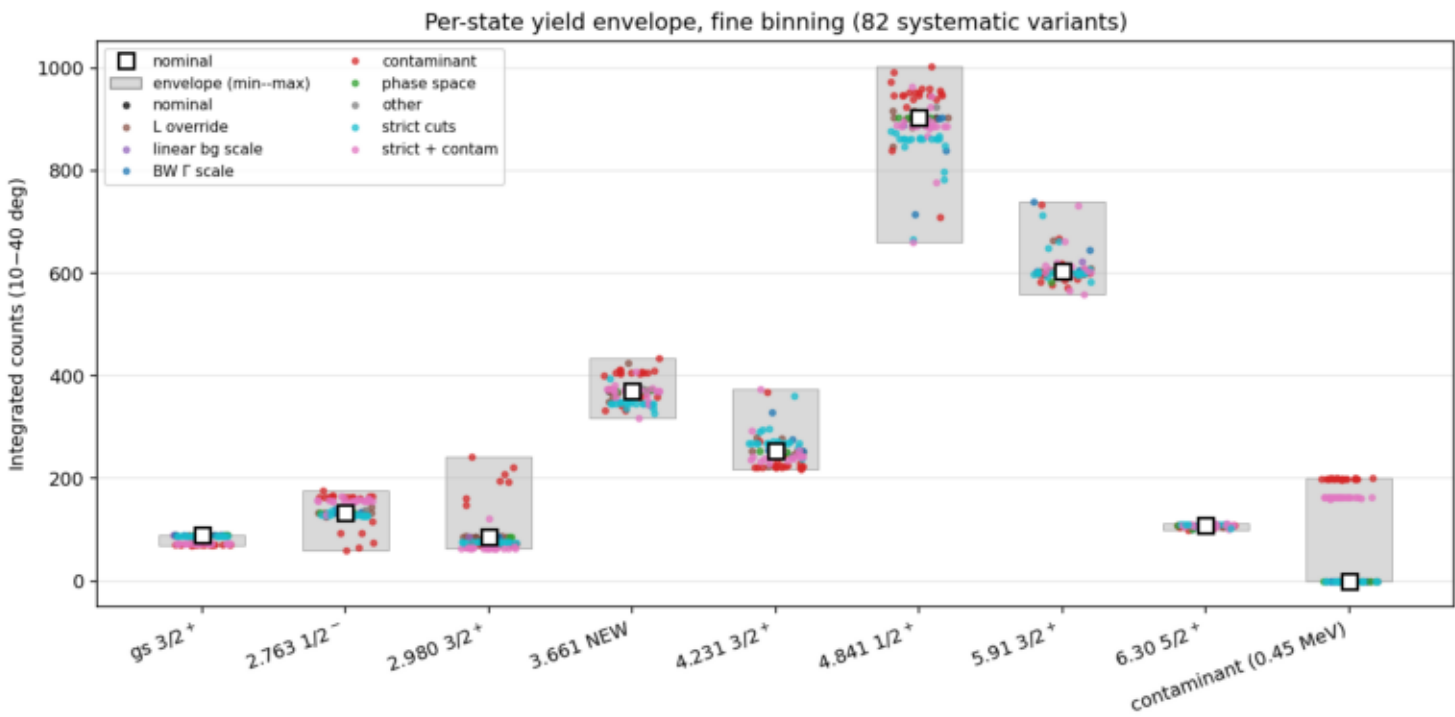
best L (corrected errors):

Ex	best L	next L

2.763	L=3 (1.41)	L=2 (1.42)
2.980	L=3 (2.73)	L=2 (3.05)
3.661	*L=2 (1.38)	L=3 (1.45)
4.231	*L=2 (0.58)	L=3 (1.38)
4.841	*L=0 (3.09)	L=1 (7.35)
5.91	L=2 (3.55)	L=0 (3.73)
6.30	L=0 (0.34)	L=3 (0.40)

- χ^2/ν refreshed after the error fix.
- All best-L picks & tie flags UNCHANGED;
 - χ^2/ν rose $\times 1.3$ – 1.5 (every L scales together).

- 4.231 $3/2^+$ / 4.841 $1/2^+$: decisive (L=2 d-wave; L=0 s-wave) — next L worse by factor $\gtrsim 2$; match the PDF.
- 3.661 (new): L=2 over L=3 (1.38 vs 1.45) — not decisive; possible sd-pf intruder.
- 2.763 $1/2^-$: three-way near-tie (L=3/2/1); AD too flat to separate.
- 2.980 $3/2^+$: no template fits well; 17.5° AD peak not reproduced; fit wants $E_0 \sim 120$ keV above the SM prior.
- 6.30 $5/2^+$: L=0 has best χ^2/ν but is FORBIDDEN ($0^+ \otimes s_{1/2} = 1/2^+$); allowed L=2/3 are tied.
- 5.91 $3/2^+$: L=2 marginal over L=0; AD poorly constrained past 30°.



Integrated-yield envelope:

state	-%	+%

g.s. 3/2+	-23	+0
2.763	-56	+32
2.980	-28	+179
3.661	-15	+17
4.231	-14	+48
4.841	-27	+11
5.91	-8	+22
6.30	-9	+3

- Largest model-dependence: 2.763 & 2.980 (contaminant degeneracy).
- 6.30 robust (<10%).

Summary

- Per-state absolute $d\sigma/d\Omega$ for 3 bound + 7 unbound ^{17}C states from $^{16}\text{C}(d,p)$ at 11.77 MeV/u.
- Absolute scale validated vs $^{16}\text{C}(d,d)$ elastic; normalization uncertainty = $\sim 12\%$ OMP spread (global).
- Error model corrected: per-point stats no longer double-counted; statistical + systematic + scale layers separated.
- L assignments robust to the fix: 4.231 (L=2) & 4.841 (L=0) decisive; 3.661/2.763/5.91 ambiguous; 2.980 & 6.30 problematic.
- Deliverable (this repo): yields, $d\sigma/d\Omega$, systematic envelope. Final L / C^2S done downstream (CDCC-grade).

Open items

- Fold the $\sim 12\%$ elastic-normalization scale into the published $d\sigma/d\Omega$ / yield bands (common factor).
- Continuum-form tests (quadratic/exp bg, PS-shape) to show the 0.45 MeV contaminant is not bg flexibility.
- Physically identify the 0.45 MeV excess (vertex_z, θ_{lab} fingerprint of a non-D₂ channel?).
- CDCC / pole-residue treatment for publication-grade absolute C^2S .

Refs: Pereira-Lopez et al., PLB 811 (2020) 135939; Movilla & Ayyad (2026); Pang et al., arXiv:1606.01507 (DA1p).

Backup: full χ^2/ν grid

L values

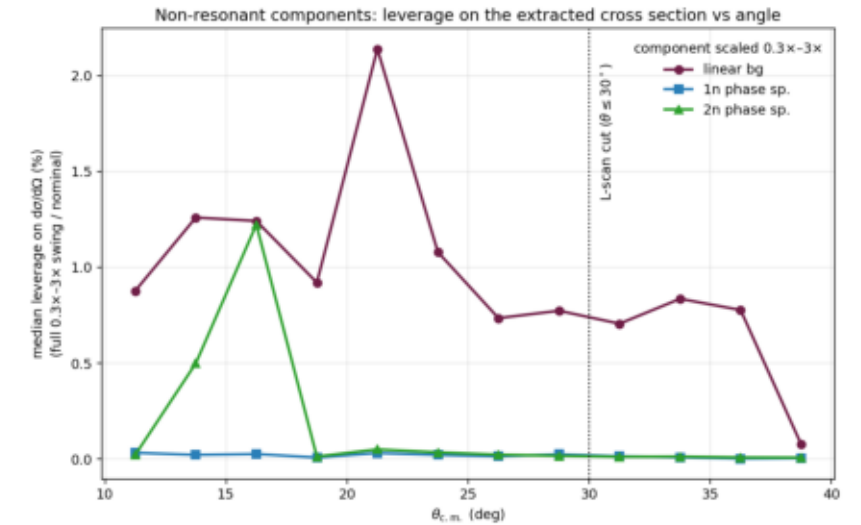
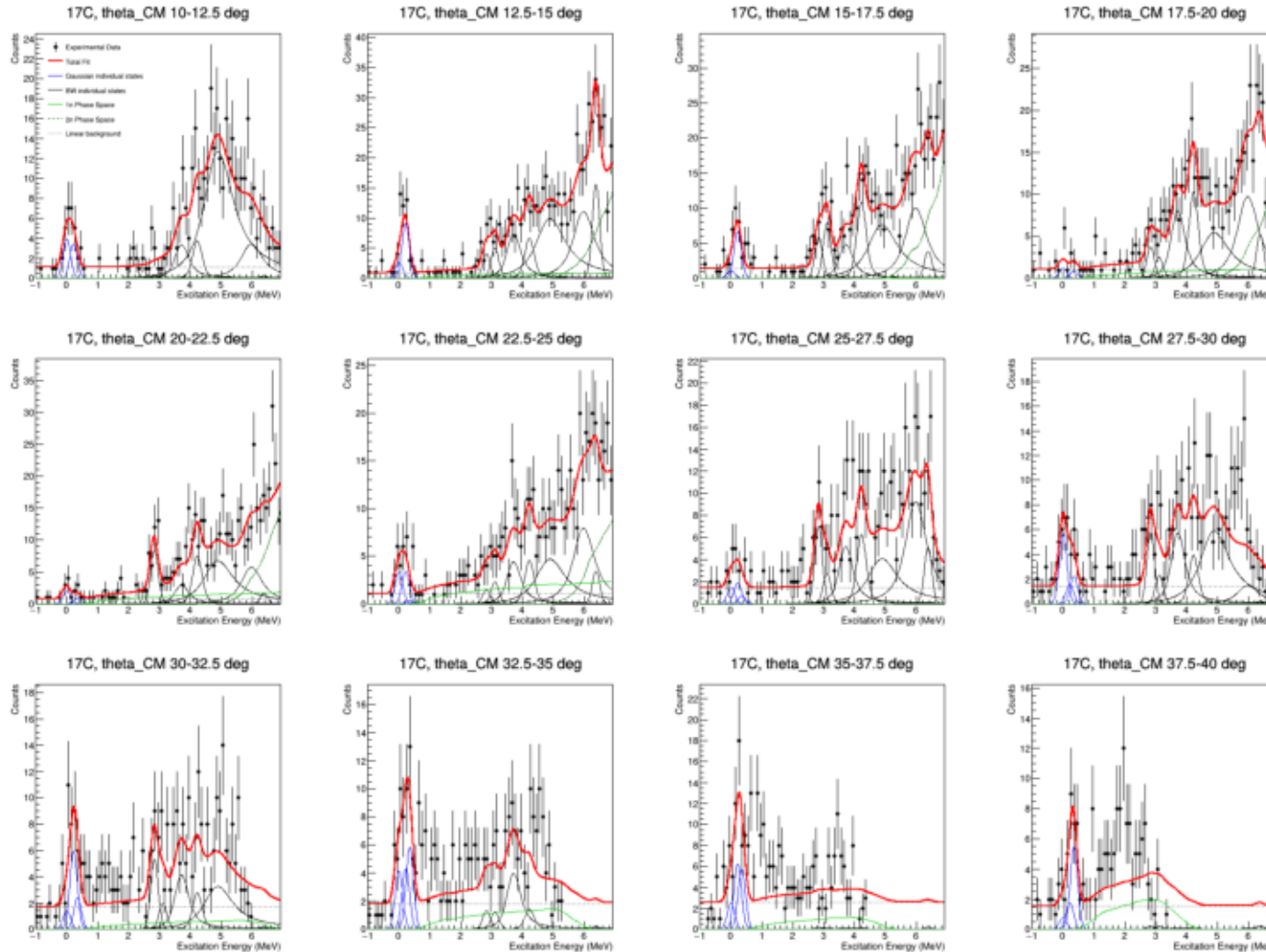
state	L=0	L=1	L=2	L=3	best

2.763 1/2-	3.27	1.61	1.42 *	1.41	L=3 (tie L=2)
2.980 3/2+	3.83	3.66	3.05 *	2.73	L=3
3.661 new	3.68	4.35 *	1.38	1.45	L=2 (tie L=3)
4.231 3/2+	2.87	2.12 *	0.58	1.38	L=2
4.841 1/2+ *	3.09	7.35	9.23	11.53	L=0
5.91 3/2+	3.73	4.25 *	3.55	5.10	L=2 (tie L=0)
6.30 5/2+ *	0.34	0.72	0.41	0.40	L=0 (tie L=3)

* = best L. Fits on theta_cm <= 30 deg, corrected stat errors (2026-05-30).
A tie = runner-up within 0.3 in chi2/nu (shapes indistinguishable at current statistics).

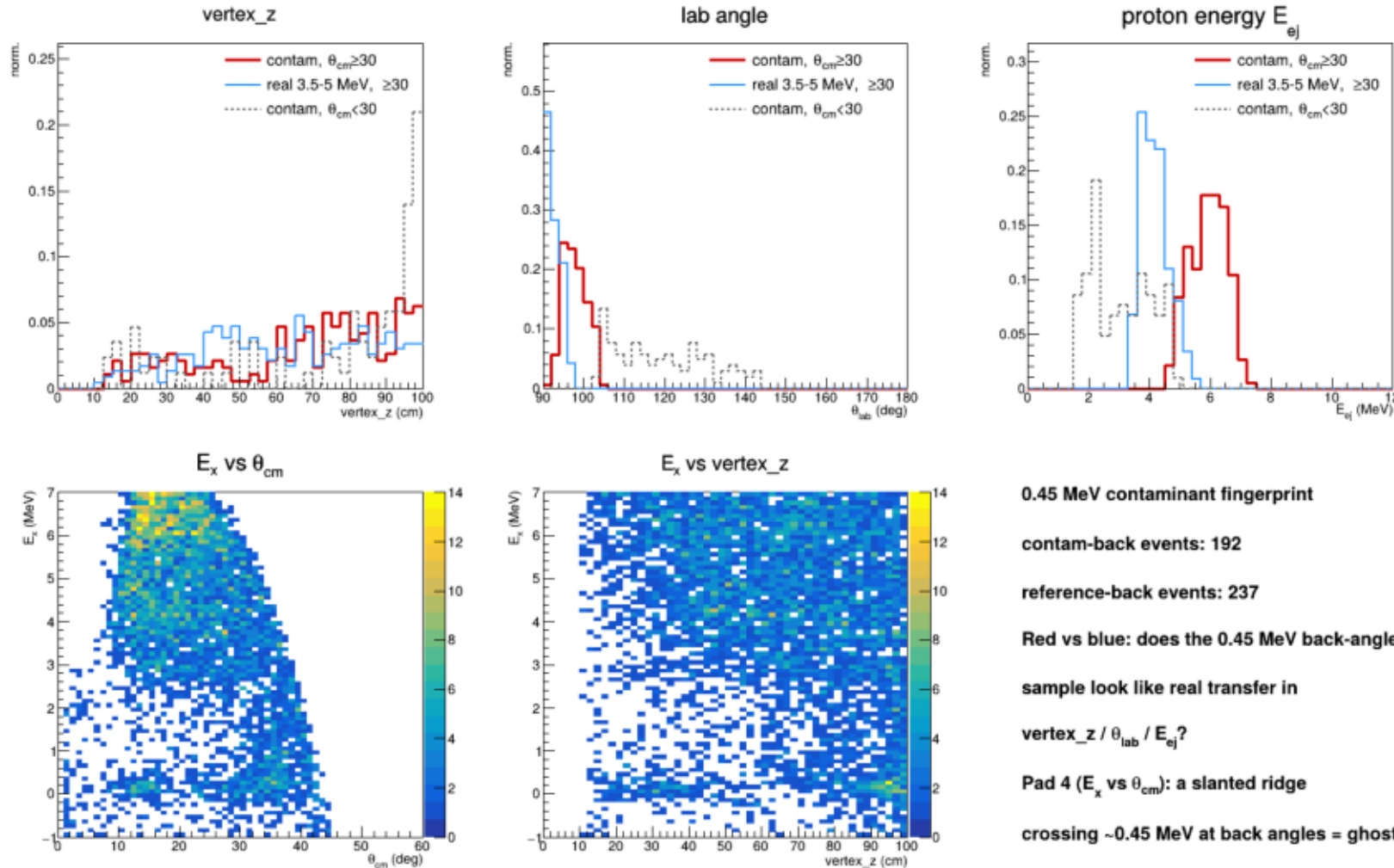
Backup: non-resonant background treatment

background



- In the fit: 1n PS + linear bg + 2n PS, amplitudes free in EVERY angular bin (PS templates rebuilt per bin).
- Left: per-bin E_x spectra — peak-dominated forward; from $\sim 30^\circ$ the transfer peaks fade and the continuum dominates.
- Right: scaling each term $0.3\times\text{--}3\times$ moves $d\sigma/d\Omega$ by $<2\%$ (bg), $\sim 0\%$ (PS) — flat in angle, no rise past 30° .
- Also in the 82-variant envelope; L-scan uses $\theta \leq 30^\circ$. Back-angle limit is stats/efficiency, not bg.

Backup: 0.45 MeV contaminant - kinematic fingerprint (inconclusive)



- Back-angle feature: $\sim 87\%$ of its fitted yield at $\theta_{cm} \geq 30^\circ$ (contam_vs_angle).
- Caveat: the E_{ej} comparison is CIRCULAR — E_x is built from E_{ej}, θ , so the window's protons must follow low- E_x kinematics; it can't separate signal from background.
- E_x -independent vertex_z: the back-angle contaminant looks like genuine transfer — no clear anomaly.
- $\Rightarrow$ this simple test does NOT flag it as background; its nature is still open.
- Next: E_x -independent discriminants — track redchisq, spatial clustering, slanted $E_x(\theta)$ ridge vs vertical band.

0.45 MeV contaminant fingerprint

contam-back events: 192

reference-back events: 237

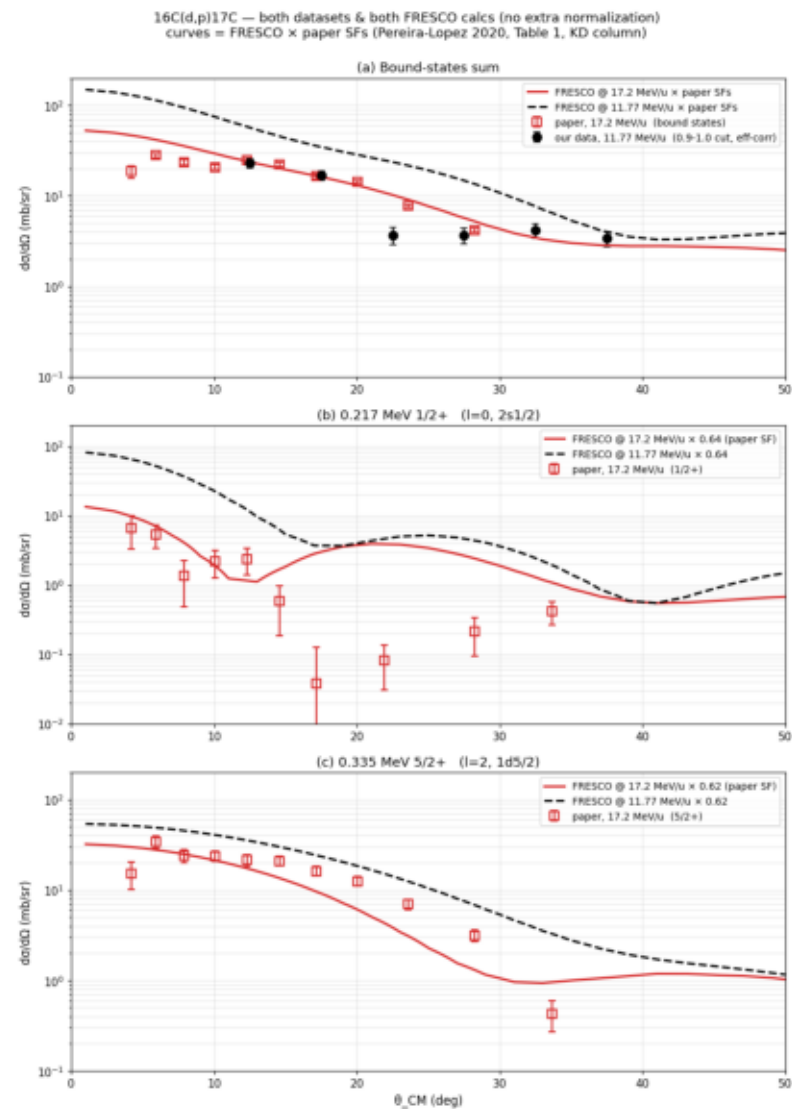
Red vs blue: does the 0.45 MeV back-angle

sample look like real transfer in

vertex_z / θ_{lab} / E_{ej} ?

Pad 4 (E_x vs θ_{cm}): a slanted ridge

crossing ~ 0.45 MeV at back angles = ghost.



Bound-state C^2S : PLB vs this work

state	l	PLB	this work

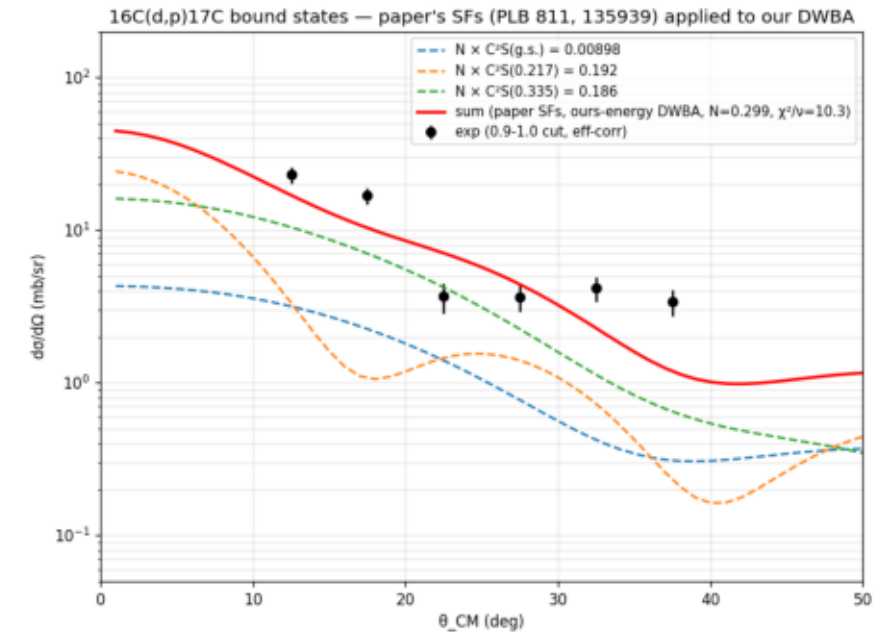
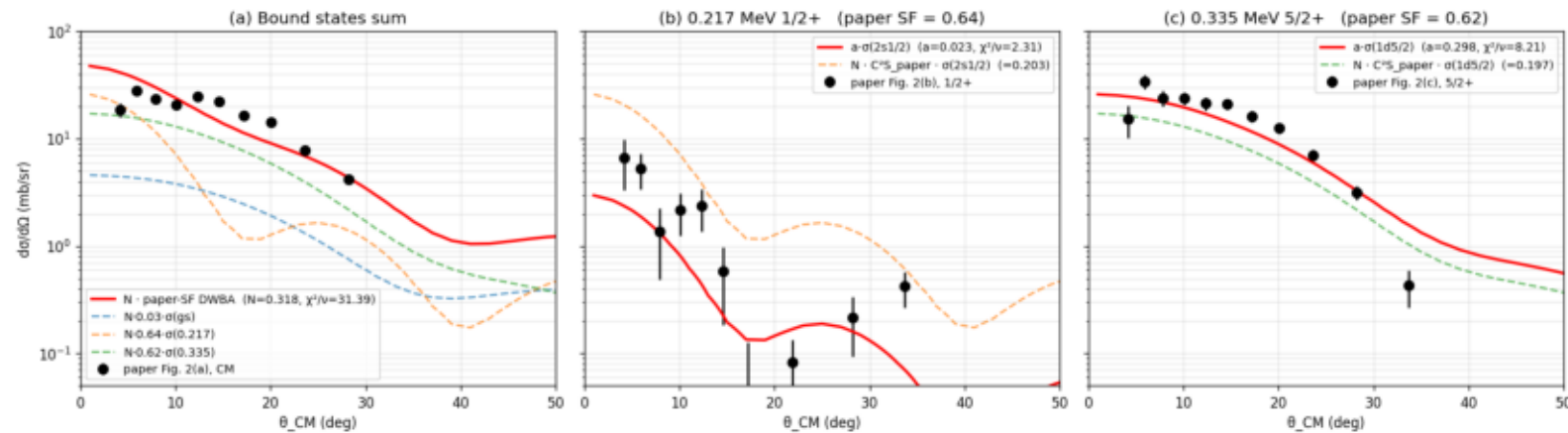
gs $3/2^+$	2	0.03	$\rightarrow 0$
0.217 $1/2^+$	0	0.64(18)	0.54(10)
0.335 $5/2^+$	2	0.62(13)	0.47-0.74

- CM-frame angular distributions (both datasets, both FRESKO energies, no extra normalization).
- Our absolute scale is validated vs the $^{16}\text{C(d,d)}$ elastic channel (same beam $\times$ target luminosity).
- $1/2^+$ and $5/2^+$ C^2S reproduced within $\sim 1\sigma$; g.s. $3/2^+$ consistent with the small published value.
- Companion analysis: repo 16Cdp17C_PLB_comparison.

PLB 811 comparison: per-state bound-state fits

bound states

$^{16}\text{C}(\text{d},\text{p})^{17}\text{C}$ — paper's CM angular distributions vs our FRESKO DWBA $\times$ paper SFs
global $N = 0.318$ (ADWA-vs-DWBA + energy difference: 17.2 vs 11.77 MeV/u)



Left: per-state $d\sigma/d\Omega$ vs FRESKO with the published spectroscopic factors (a: bound-states sum, b: 0.217 $1/2^+$, SF=0.64, c: 0.335 $5/2^+$, SF=0.62). Right: paper SFs applied to our DWBA, summed vs efficiency-corrected data.